

TOPGRADE INNOVATION

Major Project Title :

Predictive Analysis of Student Performance Using Linear Regression

Submitted by:

BINDU BS

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1. Problem Statement

Educational institutions aim to improve student outcomes by understanding the factors associated with academic performance and identifying students who may need additional support. Manual analysis of a large student dataset can be difficult and time-consuming.

This project applies data science techniques to analyze student-related information and predict a student's Grade Point Average (GPA). The model uses demographic, study-related, attendance, support, and extracurricular information to estimate future academic performance. Such predictions can help institutions identify patterns and support informed academic interventions.

2. Objective

The main objective of this project is to develop a regression-based machine learning system that predicts student GPA from student-related attributes.

The specific objectives are:

- Load and analyze a student performance dataset using Python and Pandas.
- Perform exploratory data analysis to identify patterns in GPA, study time, and absences.
- Check data quality by identifying missing values and reviewing data types.
- Preprocess the dataset by removing irrelevant and leakage-prone columns.

- Train a Linear Regression model to predict GPA.
- Evaluate the model using MAE, RMSE, and R^2 score.
- Predict GPA and performance level for a new hypothetical student.

3. Tools and Technologies

The following tools and technologies were used:

- **Python** – Main programming language used for analysis and machine learning.
- **Jupyter Notebook** – Used to perform exploratory data analysis, model development, and prediction experiments.
- **Visual Studio Code** – Used to create and execute the final Python source file.
- **Pandas** – Used to load, clean, inspect, and manipulate the dataset.
- **NumPy** – Used for numerical calculations, including RMSE calculation.
- **Matplotlib** – Used to create visualizations.
- **Seaborn** – Used to create statistical graphs and plots.
- **Scikit-learn** – Used for train-test splitting, Linear Regression modeling, and evaluation metrics.
- **Microsoft Word** – Used to prepare the project documentation and export it as PDF.

4. Dataset Description

The dataset used in this project is a Student Performance Dataset containing 2,392 student records and 15 columns. Each row represents one student and contains demographic information, study habits, attendance, academic support information, extracurricular involvement, GPA, and GradeClass.

The target variable selected for prediction is GPA, which ranges from 0.00 to 4.00. GPA is a continuous numeric value; therefore, this problem is treated as a regression problem.

The dataset includes the following columns:

- **StudentID** – Unique identifier for each student.
- **Age** – Student age.
- **Gender** – Encoded gender value.
- **Ethnicity** – Encoded ethnicity value.
- **Parental Education** – Encoded parental education level.
- **StudyTime Weekly** – Weekly study time in hours.
- **Absences** – Number of student absences.
- **Tutoring** – Indicates whether the student receives tutoring.
- **Parental Support** – Encoded level of parental support.
- **Extracurricular** – Indicates extracurricular participation.
- **Sports** – Indicates sports participation.
- **Music** – Indicates music participation.
- **Volunteering** – Indicates volunteering participation.
- **GPA** – Student Grade Point Average; selected as the prediction target.
- **GradeClass** – Grade category.

StudentID was removed because it is only an identifier and has no meaningful relationship with academic performance. GradeClass was also removed from the model features because it is related to GPA and could lead to data leakage, producing an unrealistically high prediction result.

5. Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, distribution, and relationships within the dataset.

The dataset contains 2,392 rows and 15 columns. Data-type analysis showed that all variables were numeric, with integer and floating-point data types. Missing-value analysis confirmed that the dataset contains zero missing values in all columns. Therefore, no imputation or row deletion was required.

Summary statistics showed that GPA ranges from 0.00 to 4.00, with an average GPA of approximately 1.91. Weekly study time ranges from approximately 0 to 20 hours, with an average of approximately 9.77 hours. Absences range from 0 to 29, with an average of approximately 14.54 absences.

The visual analysis showed that GPA values are distributed across the range from 0 to 4. The relationship between study time and GPA is positive but not very strong when considered alone. The strongest visible pattern was between absences and GPA: students with higher absences generally had lower GPAs.

6. Data Preprocessing

The following preprocessing steps were performed before training the model:

- The dataset was loaded using Pandas.
- Missing values were checked, and no missing values were found.
- StudentID was removed because it is only a unique identifier.
- GradeClass was removed because it is related to the target variable GPA and could cause data leakage.
- GPA was selected as the target variable.
- The remaining 12 variables were used as input features.
- The dataset was divided into 80% training data and 20% testing data using `train_test_split` with `random_state = 42`.

The training dataset contained 1,913 records, and the testing dataset contained 479 records. The test data was not used during model training and was used only to evaluate the final model.

7. Algorithm Explanation

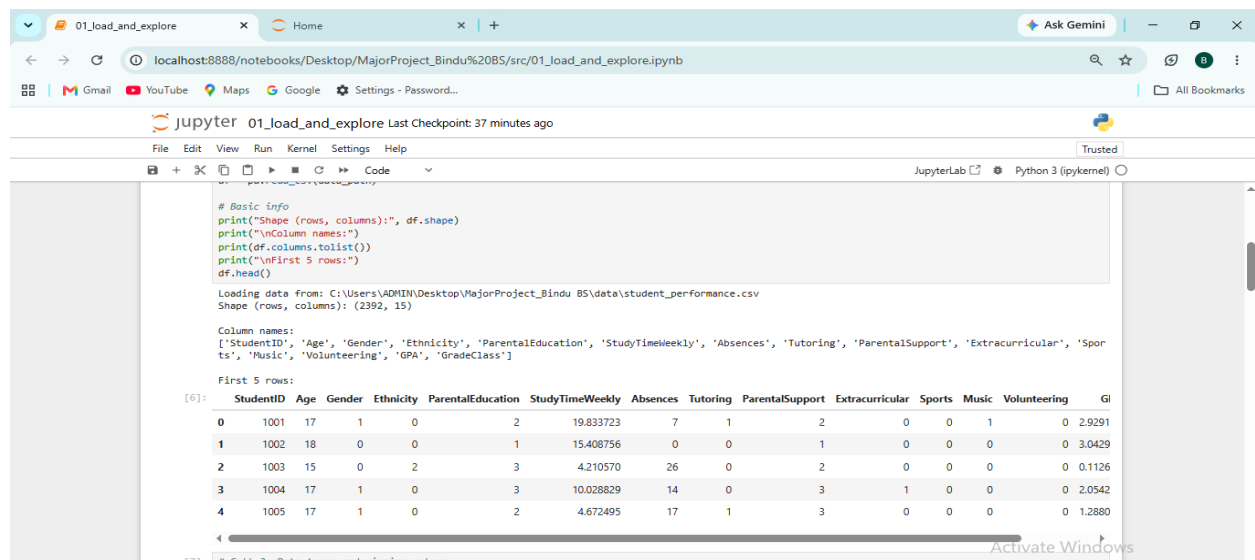
Linear Regression was selected as the machine learning algorithm because the target variable, GPA, is continuous and numeric.

Linear Regression predicts a numeric output by learning the relationship between input variables and the target variable. In this project, the model learns how variables such as study time, absences, tutoring, parental support, and extracurricular activities are associated with GPA.

The model was trained using the 80% training dataset. It then predicted GPA values for the unseen 20% test dataset. The predicted values were compared with the actual GPA values using three evaluation metrics:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between actual and predicted GPA.
- **Root Mean Squared Error (RMSE):** Measures prediction error while assigning greater importance to larger errors.
- **R-squared (R^2) Score:** Measures the proportion of variation in GPA explained by the model. A value closer to 1 indicates a better fit.

8. Screenshots



```
# Basic info
print("Shape (rows, columns):", df.shape)
print("\nColumn names:")
print(df.columns.tolist())
print("\nFirst 5 rows:")
df.head()
```

Loading data from: C:\Users\ADMIN\Desktop\MajorProject_Bindu\BS\data\student_performance.csv
Shape (rows, columns): (2392, 15)

Column names:
['StudentID', 'Age', 'Gender', 'Ethnicity', 'ParentalEducation', 'StudyTimeWeekly', 'Absences', 'Tutoring', 'ParentalSupport', 'Extracurricular', 'Sports', 'Music', 'Volunteering', 'GPA', 'GradeClass']

First 5 rows:

	StudentID	Age	Gender	Ethnicity	ParentalEducation	StudyTimeWeekly	Absences	Tutoring	ParentalSupport	Extracurricular	Sports	Music	Volunteering	GPA
0	1001	17	1	0	2	19.833723	7	1	2	0	0	1	0	2.9291
1	1002	18	0	0	1	15.408756	0	0	1	0	0	0	0	3.0429
2	1003	15	0	2	3	4.210570	26	0	2	0	0	0	0	0.1126
3	1004	17	1	0	3	10.028829	14	0	3	1	0	0	0	2.0542
4	1005	17	1	0	2	4.672495	17	1	3	0	0	0	0	1.2880

Figure 1: Preview of the Student Performance Dataset

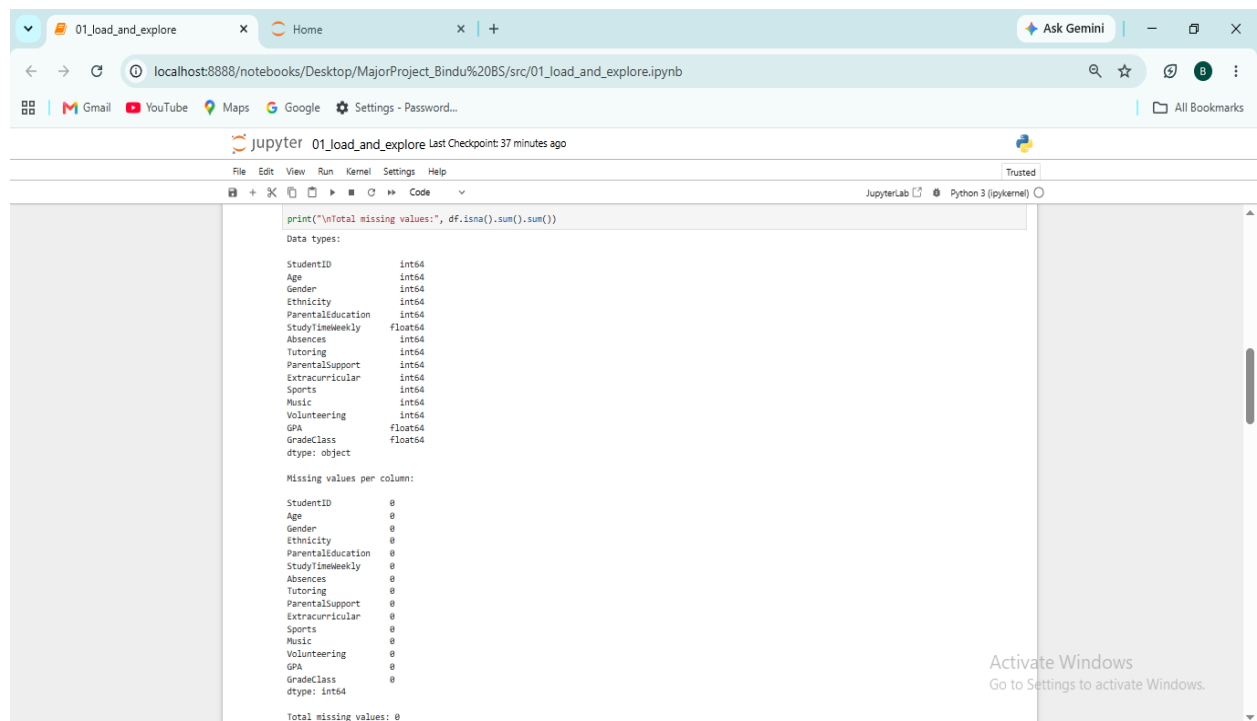


Figure 2: Data Types and Missing-Value Analysis

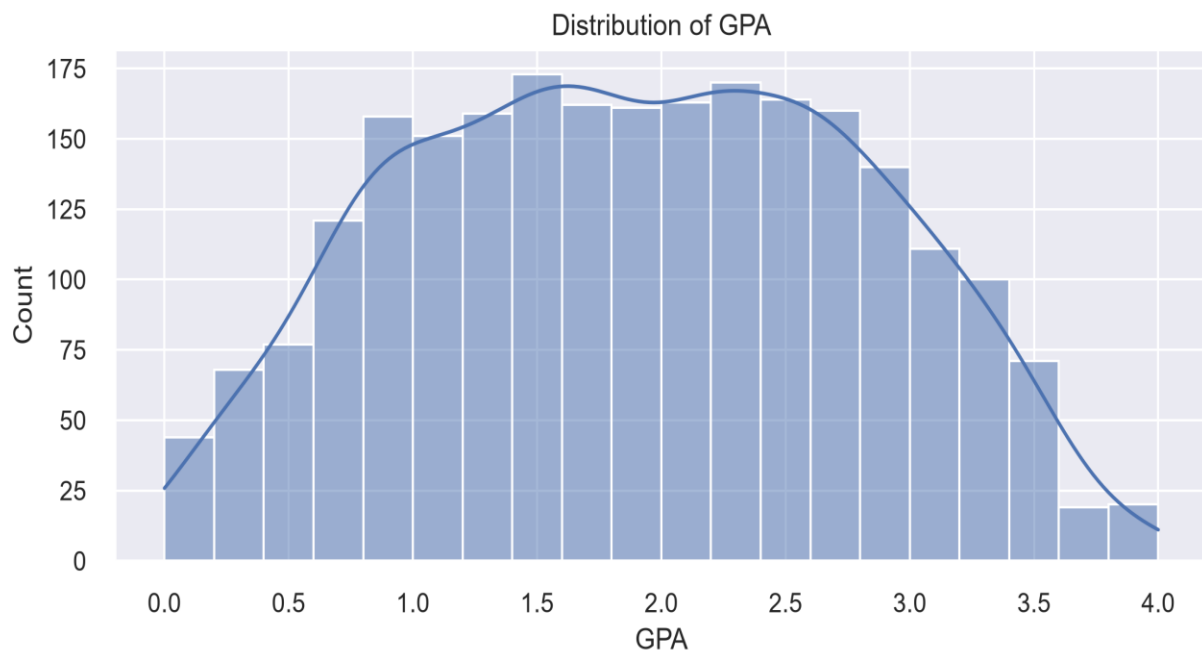


Figure 3: Distribution of Student GPA

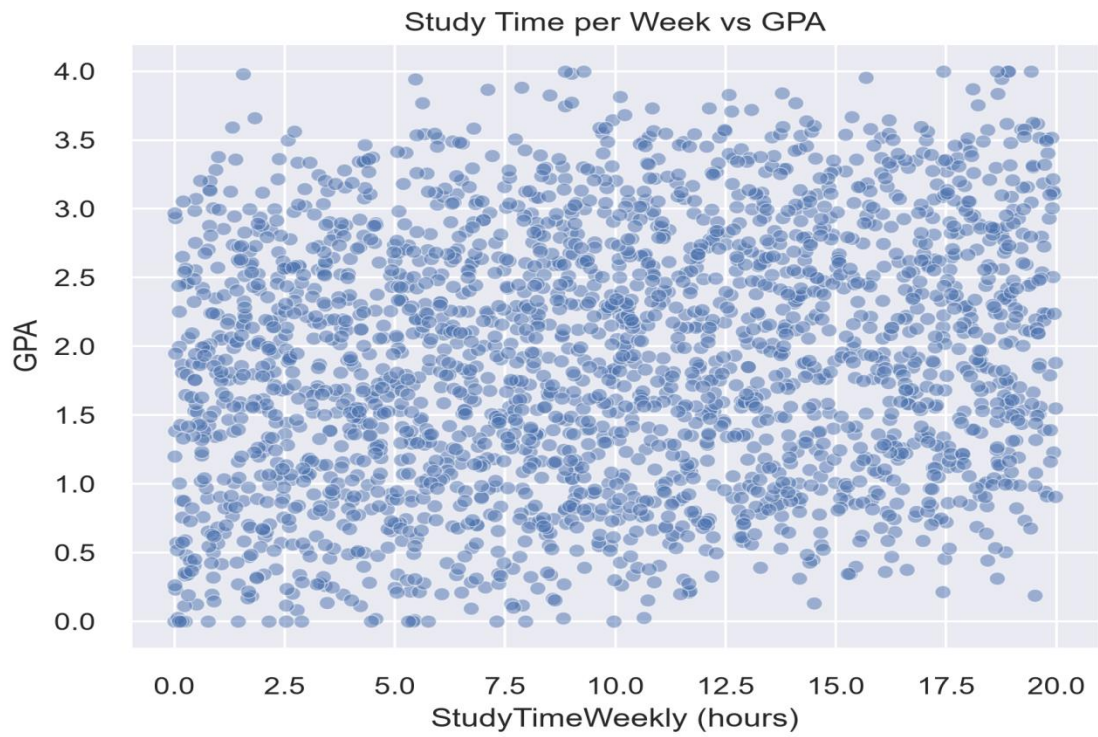


Figure 4: Relationship Between Weekly Study Time and GPA

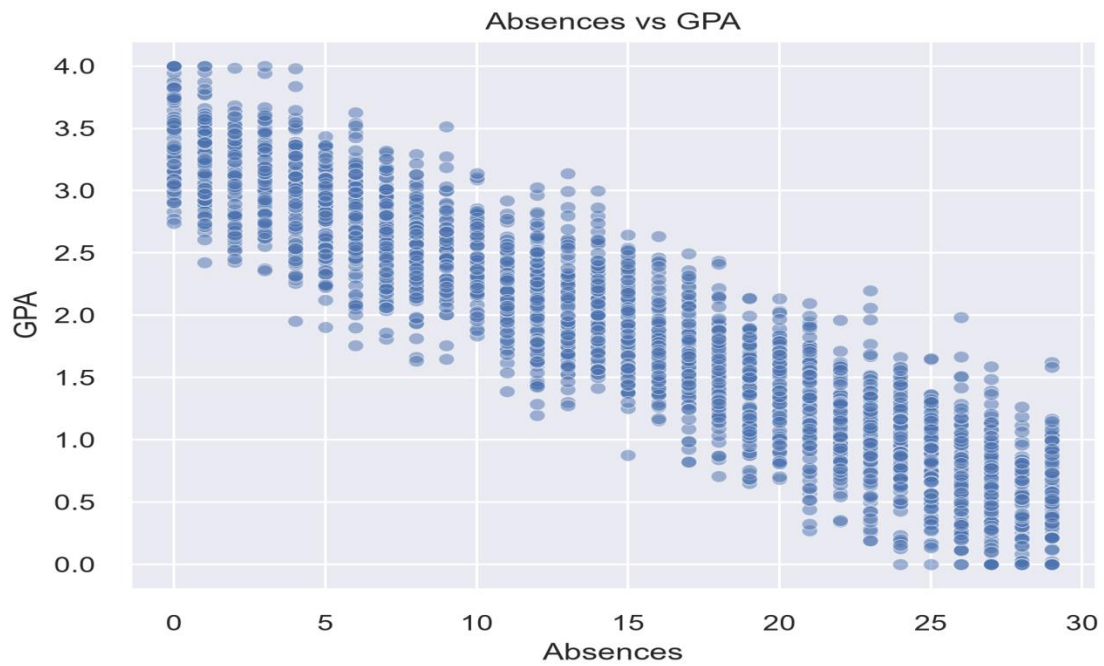


Figure 5: Relationship Between Absences and GPA

```
[7]: # Make predictions and evaluate model performance

# Predict GPA for the unseen test data
y_pred = model.predict(X_test)

# Calculate evaluation metrics
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("Model Performance on Test Data")
print("-" * 35)
print(f"Mean Absolute Error (MAE): {mae:.3f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.3f}")
print(f"R-squared (R²) Score: {r2:.3f}")

Model Performance on Test Data
-----
Mean Absolute Error (MAE): 0.155
Root Mean Squared Error (RMSE): 0.197
R-squared (R²) Score: 0.953
```

Figure 6: Linear Regression Model Performance on Test Data

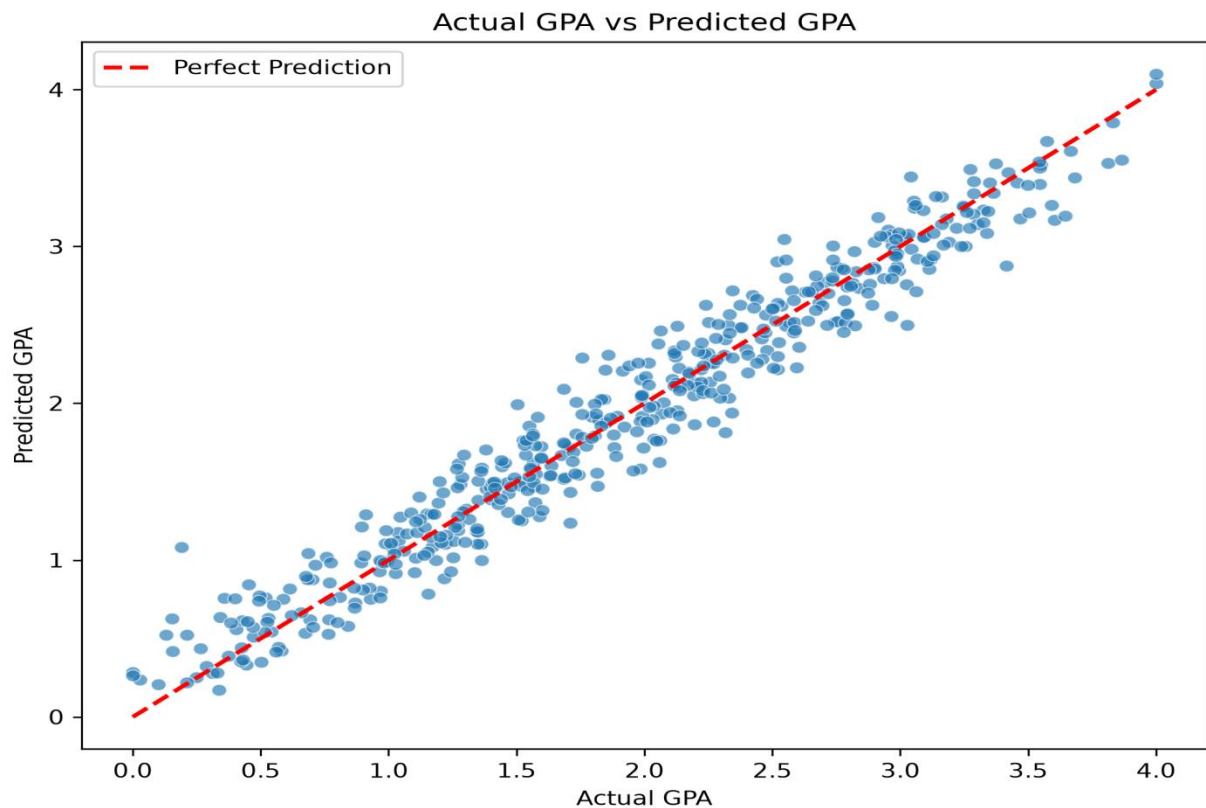


Figure 7: Actual GPA Versus Predicted GPA

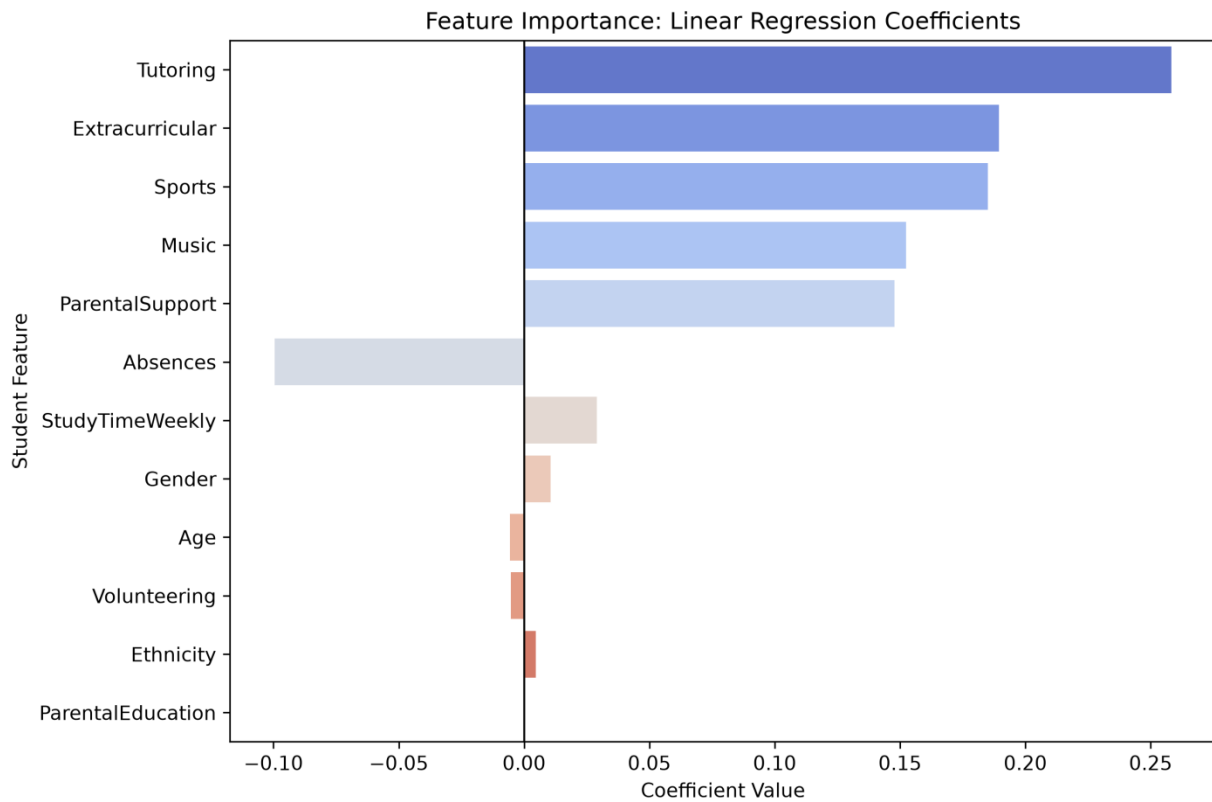


Figure 8: Feature Importance Based on Linear Regression Coefficients

```
predicted_gpa = model.predict(new_student)[0]

print("New Student GPA Prediction")
print("-" * 35)
print(f"Predicted GPA: {predicted_gpa:.2f} out of 4.00")

if predicted_gpa >= 3.0:
    performance_level = "Excellent"
elif predicted_gpa >= 2.0:
    performance_level = "Good"
elif predicted_gpa >= 1.0:
    performance_level = "Average"
else:
    performance_level = "Needs Improvement"

print(f"Predicted Performance Level: {performance_level}")

New Student GPA Prediction
-----
Predicted GPA: 3.45 out of 4.00
Predicted Performance Level: Excellent
```

Figure 9: GPA Prediction for a New Student

9. Results and Insights

The Linear Regression model was evaluated using unseen test data. The results obtained are shown below:

- Mean Absolute Error (MAE): 0.155
- Root Mean Squared Error (RMSE): 0.197
- R-squared (R^2) Score: 0.953

The MAE value of 0.155 indicates that, on average, the predicted GPA differs from the actual GPA by approximately 0.16 GPA points. The RMSE value of 0.197 indicates that larger prediction errors are also relatively low. The R^2 score of 0.953 means that the model explains approximately 95.3% of the variation in GPA on the unseen test data.

This is a strong model result for this dataset. However, the result should be interpreted as performance on this particular dataset and not as a guarantee that the model will perform identically for all student populations.

The project produced the following key insights:

- Tutoring had the strongest positive coefficient (0.258), indicating a positive association with predicted GPA.
- Extracurricular participation, sports participation, music participation, and parental support were also positively associated with predicted GPA.
- Absences had a negative coefficient (-0.100), indicating that higher absence counts were associated with lower predicted GPA.
- StudyTimeWeekly had a positive coefficient (0.029), indicating that additional study time was associated with a modest increase in predicted GPA.
- The Actual GPA versus Predicted GPA graph showed that most predicted values were close to the perfect-prediction reference line, supporting the strong R^2 score.

- A hypothetical student with 12 weekly study hours, 5 absences, tutoring support, and parental support received a predicted GPA of 3.45 out of 4.00. The predicted performance level was Excellent.

These findings show that attendance, academic support, study habits, and student engagement are useful variables for predicting academic performance in this dataset. The model identifies associations in the available data and does not prove direct cause-and-effect relationships.

10. Conclusion

This project successfully developed a student performance prediction system using Python, Pandas, Scikit-learn, Matplotlib, and Linear Regression.

The dataset was analyzed through exploratory data analysis, checked for missing values, and preprocessed by removing irrelevant and leakage-prone columns. A Linear Regression model was trained on 80% of the data and tested on the remaining 20%.

The model achieved an MAE of 0.155, an RMSE of 0.197, and an R^2 score of 0.953 on the test data. It also successfully predicted the GPA of a new hypothetical student as 3.45 out of 4.00, categorized as Excellent.

The project demonstrates how data science can help educational institutions analyze factors associated with academic performance and support data-informed student interventions.

11. Future Scope

- Compare Linear Regression with advanced models such as Random Forest Regressor and Gradient Boosting Regressor to improve prediction performance.
- Apply cross-validation to test the model's stability across different data splits.
- Use real institutional data with proper privacy and ethical safeguards.
- Develop a simple web application or dashboard to predict GPA and identify students who may need academic support.